Git and GitHub

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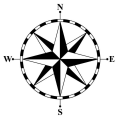
Plan for Session



- Git Introduction
- Single Project Repository
- Git Source Code Control
- Git Change Flow
- File States in Git
- Git Tools
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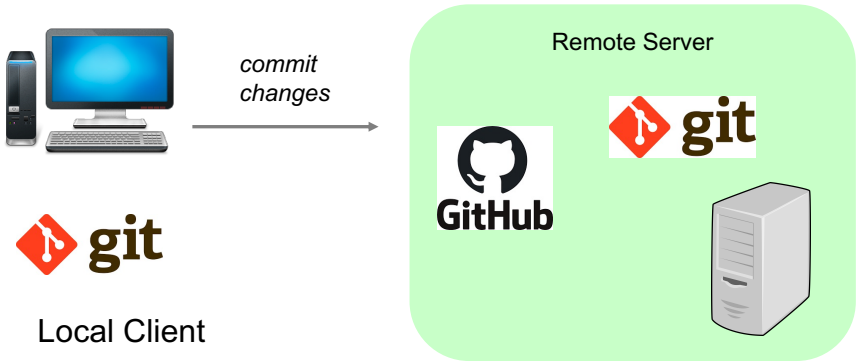
- Adding Files in Git
- Working with a Remote Repository
- GitHub
- Branching & Pull Requests

Git Introduction

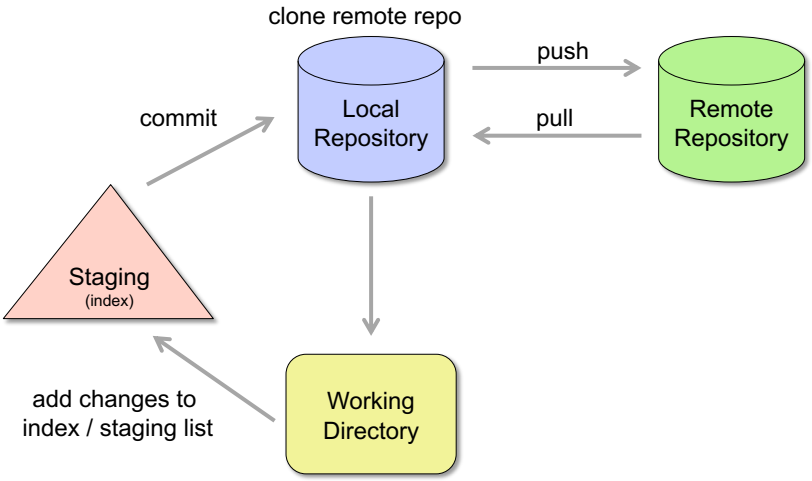
- Git is a
  - Source Code Configuration Management system
  - aka SCCM or SCM system
- Shared repository of code
  - to allow teams to work on codebase of a project
- Track software changes
  - so that those changes can be undone if necessary
- Tag releases
  - know exactly what code made it into a release
  - useful when different versions are needed for different clients

Git Introduction

- Develop code local store centrally



Git Source Code Control



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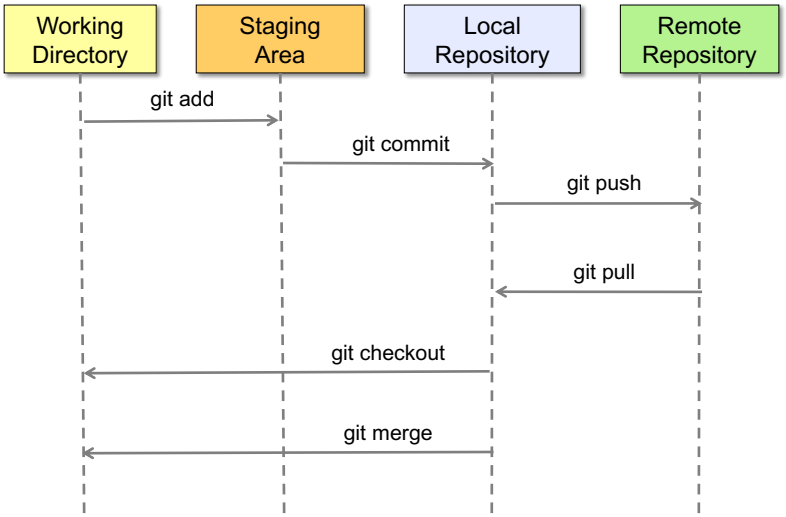
Git Change Flow



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File States in Git

- Files can be in one of four states

State	Description
untracked	file is not tracked by Git
staged	file is included in list of changes to be committed
tracked	file has been staged and then committed, Git tracks changes to this file
modified	file has changed but not yet been staged

Git Tools

- Command Line Tools
  - basic command line interface for Git
- GitHub
  - web based interface to Git repos
  - supports Pull Requests (PRs) and other workflow options
  - can support reviews
  - diff tools etc.
- IDEs Git plugins / support
  - Git support in PyCharm
  - Git support also in Visual Studio Code etc.

Command Line Git

- Available for Mac, Windows and Linux
- Can be used to create a new project
  - can be useful for projects exploring how to solve problems
    - e.g. commit at regular intervals as you test new functionality
- Need to initialize a new git project

```
$ mkdir my_project
$ cd my_project
$ git init
Initialized empty Git repository in my_project/.git/
$
```


Command Line Git

- Can also clone an existing repository

```
$ git clone https://github.com/johnehunt/javascript-intro.git
Cloning into 'javascript-intro'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 7152 (delta 0), reused 4 (delta 0), pack-reused 7144
Receiving objects: 100% (7152/7152), 27.51 MiB | 7.71 MiB/s, done.
Resolving deltas: 100% (2495/2495), done.
Checking connectivity... done
$
```

  - can now make changes to this repository
  - add them
  - commit them
  - push them to the GitHub hosted repo

Adding Files in Git

- Need to stage changes
  - adds them to the list of changed files

```
$ cat > jjh-readme.txt
Johns readme

$ git add jjh-readme.txt
```

- Commit inserts the staged changes into the local repository

```
$ git commit -m 'My first commit'
[master (root-commit) 6b0dfe6] My first commit
 1 file changed, 1 insertion(+)
 create mode 100644 jjh-readme.txt
$
```


Working with Remote Repository

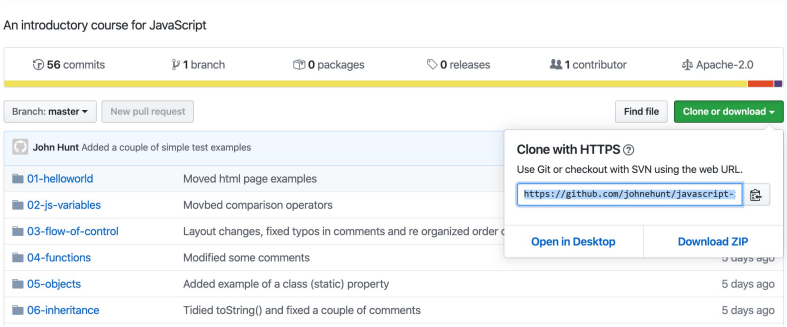
- Use the `git push` command to update the remote repository

```
$ git push
Username for 'https://github.com': johnehunt
Password for 'https://johnehunt@github.com': ******
Counting objects: 3, done.
Writing objects: 100% (3/3), 234 bytes | 0 bytes/s, done.
Total 3 (delta 0), reused 0 (delta 0)
To https://github.com/johnehunt/playarea.git
 * [new branch]      main -> main
$
```

  - may need to specify the branch
    - e.g. git push main

GitHub Interface

- Can use web interface to see comments, commits, and get clone Url information



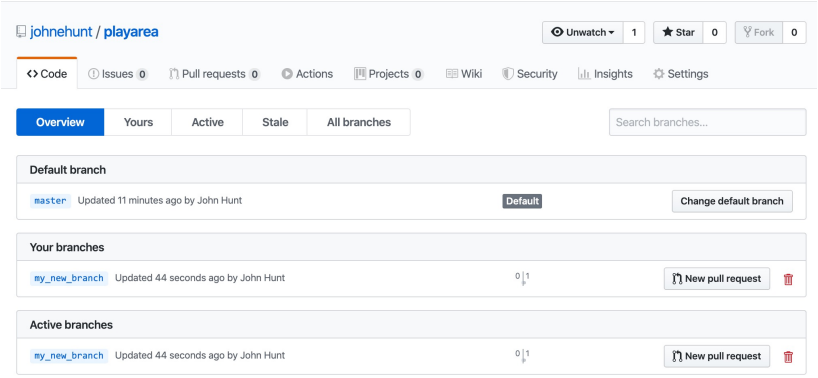
Branching

- Typically do development in a branch
  - Then merge your changes back into master/main branch
  - Typically via a Pull Request (or PR)
    - which has a second developer review the changes
    - and then pull those changes into the main branch
- Create branch using `git checkout –b <branch-name>`

```
$ git branch
* master
$ git checkout -b my_new_branch
Switched to a new branch 'my_new_branch'
$ git branch
  master
* my_new_branch
$
```

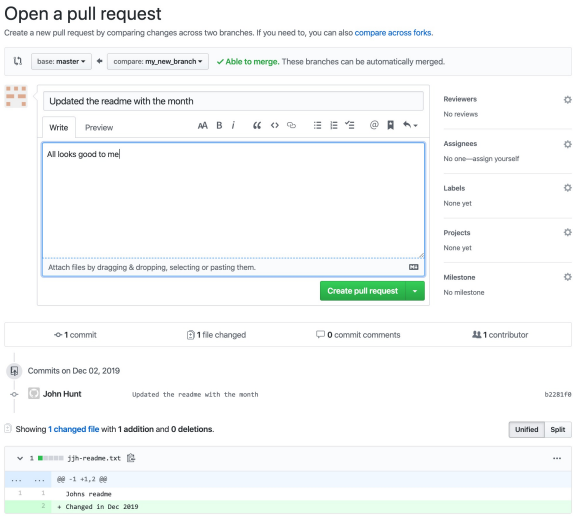
Branching

- Master / main branch and my_new_branch



Branching & Pull Requests

- Creating a PR
- Used to request changes are merged from branch to main or dev
- Another developer reviews and then *pulls* the changes into another branch



Git Terminology Quick Reference

Term	Description
HEAD	Symbolic reference to the latest version of the current branch
branch	reference to specific state of files in the working directory
staging area	Stores changes before they are committed (a change list) also called index
add	Add changes to the staging area
commit	Apply changes to the *local* repository
origin	The default upstream (remote) repository
push	Send changes from the local to the remote repository
pull	Retrieve changes from a remote repository and apply them to the local repository
pull request	Request to merge two branches following review and approval process

Useful References

- Git Home Page
  - http://git-scm.com
  - http://git-scm/doc highly recommended
- Git Glossary
  - https://www.kernel.org/pub/software/scm/git/docs/gitglossary.html
- 10 Git Commands Every Developer should Know
  - https://www.freecodecamp.org/news/10-important-git-commands-that-every-developer-should-know
- Others
  - http://www.kernel.org/pub/software/scm/git/docs
  - http://progit.org/book Pro Git Book

Questions?

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